

内容提要

- UCSC Genome Browser 简介
- Genome Browser的使用
- Table Browser 的使用
- Download genome annotation



UCSC Genome browser 简介

- 维护机构: University of California Santa Cruz (UCSC)
- 内容:
 - 许多物种的参考基因组序列及正在分析拼接中的草图基因组版本
 - 提供到ENCODE 数据库 (2003 to 2012) 以及 Neandertal计划的链接。
 - 展示了世界范围内对基因组的各种注释, 可通过网页浏览、拖动、按自己的方式展示
 - 非常多的生物信息学分析工具
 - Gene Sorter
 - Blat
 - Table Browser
 - VisiGene
 - Genome Graphs



UNIVERSITY OF CALIFORNIA
SANTA CRUZ



Genome Browser Gateway

UCSC Genome browser 简介

- 给浏览基因组数据提供了可靠、迅速、可视化的方式
- 数据来源：

一半的注释信息是UCSC通过来自公开的序列数据计算得出，另外一半来自世界各地的科学工作者。用户可以自行上传track

- 本身并不下任何结论，只是收集各种相关信息供用户参考
- 支持数据库检索和序列相似性搜索



关键在于用户对数据的解读！！！！

第二堂课的问题

- 如何下载人类全基因组的参考序列
- 如何下载H3K27Ac在K562细胞中的信号
- 如何看某一特定区域的基因组结构信息

内容提要

- UCSC Genome Browser 简介
- **Genome Browser**的使用：
 - 1) 页面布局介绍
 - 2) 如何有效检索
 - 3) 结果的显示和有效的调整

首页界面



UCSC Genome Bioinformatics

<http://genome.ucsc.edu/>

[Genomes](#)[Genome Browser](#)[Tools](#)[Mirrors](#)[Downloads](#)[My Data](#)[Help](#)[About Us](#)

[Genome Browser](#)[Blat](#)[Table Browser](#)[Gene Sorter](#)[In Silico PCR](#)[Genome Graphs](#)[Galaxy](#)[VisiGene](#)[Utilities](#)[Downloads](#)[Release Log](#)[Custom Tracks](#)

About the UCSC Genome Bioinformatics Site

Welcome to the UCSC Genome Browser website. This site contains the reference sequence and working draft assemblies for a large collection of genomes. It also provides portals to [ENCODE](#) data at UCSC (2003 to 2012) and to the [Neandertal](#) project. You may download or purchase the Genome Browser source code, or the Genome Browser in a Box (GBiB) at our [online store](#).

We encourage you to explore these sequences with our tools. The [Genome Browser](#) zooms and scrolls over chromosomes, showing the work of annotators worldwide. The [Gene Sorter](#) shows expression, homology and other information on groups of genes that can be related in many ways. [Blat](#) quickly maps your sequence to the genome. The [Table Browser](#) provides convenient access to the underlying database. [VisiGene](#) lets you browse through a large collection of *in situ* mouse and frog images to examine expression patterns. [Genome Graphs](#) allows you to upload and display genome-wide data sets.

The UCSC Genome Browser is developed and maintained by the Genome Bioinformatics Group, a cross-departmental team within the [UC Santa Cruz Genomics Institute](#) at the University of California Santa Cruz ([UCSC](#)). If you have feedback or questions concerning the tools or data on this website, feel free to contact us on our [public mailing list](#). The Genome Browser is for research use only. Not intended for clinical use.

The Genome Browser project team relies on public funding to support our work. Donations are welcome -- we have many more ideas than our funding supports! If you have ideas, drop a comment in our [suggestion box](#).

[DONATE NOW](#)

News

[News Archives](#)

To receive announcements of new genome assembly releases, new software features, updates and training seminars by email, subscribe to the [genome-announce](#) mailing list. Please see our [blog](#) for posts about Genome Browser tools, features, projects and more.

1 June 2016 - New Density Graph Feature!

navigation bar

网站简介部分

新闻通知

Open Genome browser gateway



UCSC Genome Bioinformatics

Home Genomes Genome Browser Tools Mirrors Downloads My Data Help About Us

Genome Browser

Blat
Table Browser
Gene Sorter
In Silico PCR
Genome Graphs
Galaxy
VisiGene
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Release Log

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News [News Archives](#) ►

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Old Version of Genome browser

Human (*Homo sapiens*) Genome Browser Gateway

The UCSF Genome Browser was created by the [Genome Bioinformatics Group of UC Santa Cruz](#).
Software Copyright (c) The Regents of the University of California. All rights reserved.

物种分类 物种 版本 检索词

group genome assembly position search term

Mammal Mammal Vertebrate Deuterostome Insect Nematode Other Viruses

Human Mouse Mouse Alpaca Armadillo Baboon Bushbaby Cat Chimp Chinese hamster Cow Dog Dolphin Elephant Ferret Gibbon Gorilla Guinea pig Hedgehog Horse Kangaroo rat

Feb. 2009 (GRCh37/hg19) chr18:77,254,658-77,254,669 enter position, gene symbol or search terms submit

track search add custom tracks track hubs configure tracks and display

Dec. 2013 (GRCh38/hg38) Dec. 2013 (GRCh38/hg38) Feb. 2009 (GRCh37/hg19) Mar. 2006 (NCBI36/hg18) May 2004 (NCBI35/hg17) July 2003 (NCBI34/hg16)

More on-site workshops available!

10 May 2016 - New Gateway Page!

The UCSC Genome Browser team is proud to announce a newly redesigned [Genome Browser Gateway](#) page. The Gateway retains its original functionality as a central access point for all genome assemblies available on our site, while sporting several helpful new features and updates:

- Autocomplete searching for any genome browser, genome version, or public hub
- Species browsing through a scrollable visual "tree" menu based on phylogenetic order
- Quick access shortcuts to popular browsers
- New style and color scheme

The screenshot shows the UCSC Genome Browser Gateway interface. At the top is the University of California Santa Cruz logo and the title "Genome Browser Gateway". Below this is a navigation bar with links: Genomes, Genome Browser, Tools, Mirrors, Downloads, My Data, Help, and About Us. The main content area is divided into two sections: "Browse/Select Species" and "Find Position".

Browse/Select Species

POPULAR SPECIES

Human, Mouse, Rat, Fruitfly, Worm, Yeast

Enter species or common name

REPRESENTED SPECIES

A phylogenetic tree is shown on the left, with a vertical bar indicating the selected species. The tree includes Human, Chimp, Bonobo, Gorilla, Orangutan, Gibbon, Crab-eating macaque, Rhesus, Baboon (anubis), Baboon (hamadryas), Marmoset, Squirrel monkey, Tarsier, Mouse lemur, Bushbaby, Mouse, Rat, Chinese hamster, Kangaroo rat, and Squirrel.

Find Position

Human Assembly
Dec. 2013 (GRCh38/hg38)

Position/Search Term
Enter position, gene symbol or search terms
Current position: chr9:133,252,000-133,280,861

GO

Human Genome Browser - hg38 assembly [view sequences](#)

UCSC Genome Browser assembly ID: hg38
Sequencing/Assembly provider ID: GRCh38 Genome Reference Consortium Human Reference 38 (GCA_000001405.15)
Assembly date: Dec. 2013
GenBank accession ID: GCA_000001305.2
NCBI Genome information: NCBI genome/51 (Homo sapiens)
NCBI Assembly information: NCBI assembly/883148 (GRCh38/GCA_000001405.15)
BioProject information: NCBI Bioproject: 31257

Search the assembly:

- **By position or search term:** Use the "position or search term" box to find areas of the genome associated with many different attributes, such as a specific chromosomal coordinate range; mRNA, EST, or STS marker names; or keywords from the GenBank description of an mRNA. [More information](#), including sample queries.
- **By gene name:** Type a gene name into the "search term" box, choose your gene from the drop-down list, then press "submit" to go directly to the assembly location associated with that gene. [More information](#).
- **By track type:** Click the "track search" button to find Genome Browser tracks that match specific selection criteria. [More information](#).

Download sequence and annotation data:

- [Using rsync](#) (recommended)
- [Using FTP](#)
- [Using HTTP](#)

New Version !

<http://genome.ucsc.edu/cgi-bin/hgGateway>

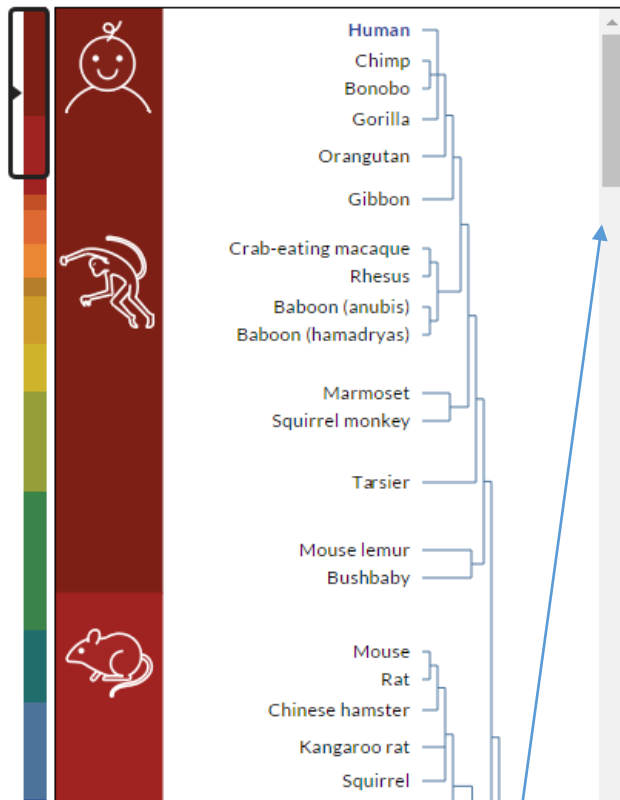
Browse/Select Species

POPULAR SPECIES



Enter species or common name

REPRESENTED SPECIES



Find Position

Human Assembly

Dec. 2013 (GRCh38/hg38)

GO

Position/Search Term

Enter position, gene symbol or search terms

Current position: chr9:133,252,000-133,280,861

Human Genome Browser - hg38 assembly

view sequences

UCSC Genome Browser assembly ID: hg38

Sequencing/Assembly provider ID: GRCh38 Genome Reference Consortium Human Reference 38 (GCA_000001405.15)

Assembly date: Dec. 2013

GenBank accession ID: GCA_000001305.2

NCBI Genome information: NCBI genome/51 (Homo sapiens)

NCBI Assembly information: NCBI assembly/883148 (GRCh38/GCA_000001405.15)

BioProject information: NCBI Bioproject: 31257



Homo sapiens
(Graphic courtesy of CBSE)

Search the assembly:

- **By position or search term:** Use the "position or search term" box to find areas of the genome associated with many different attributes, such as a specific chromosomal coordinate range; mRNA, EST, or STS marker names; or keywords from the GenBank description of an mRNA. [More information](#), including sample queries.
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- **By track type:** Click the "track search" button to find Genome Browser tracks that match specific selection criteria. [More information](#).

Download sequence and annotation data:

- Using **rsync** (recommended)
- Using **FTP**
- Using **UTTD**

基因组信息简介

基因组序列及注释信息下载

可上下拖动

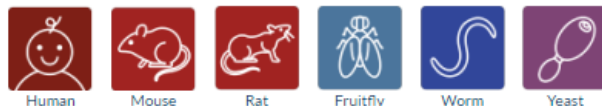
UCSC Genome Browser 收录的物种类型



- 哺乳动物（Mammals）： 52
- 脊椎动物（Vertebrates）： 19
- 后口动物（DEUTEROSTOMES）： 3
- 昆虫（Insects）： 13
- 线虫（NEMATODES）： 6
- 其他： 2（Sea hare, yeast）
- 病毒： 1（Ebola）

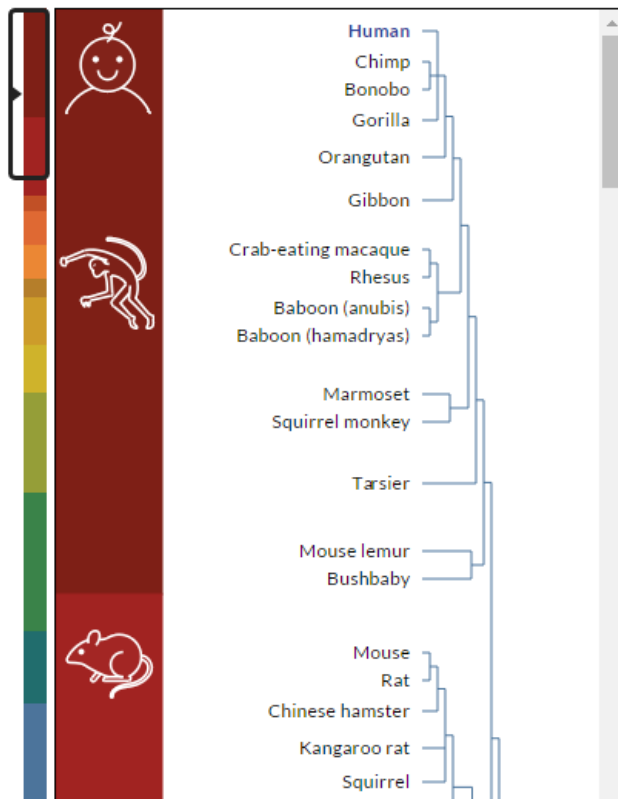
Browse/Select Species

POPULAR SPECIES



Enter species or common name

REPRESENTED SPECIES



Find Position

Human Assembly

Dec. 2013 (GRCh38/hg38)

Position/Search Term

Enter position, gene symbol or search terms

Current position: chr9:133,252,000-133,280,861



Human Genome Browser - hg38 assembly

[view sequences](#)

UCSC Genome Browser assembly ID: hg38

Sequencing/Assembly provider ID: GRCh38 Genome Reference Consortium Human Reference 38 (GCA_000001405.15)

Assembly date: Dec. 2013

GenBank accession ID: GCA_000001305.2

NCBI Genome information: NCBI genome/51 (Homo sapiens)

NCBI Assembly information: NCBI assembly/883148 (GRCh38/GCA_000001405.15)

BioProject information: NCBI Bioproject: 31257

基因组信息简介



Homo sapiens
(Graphic courtesy of CBSE)

Search the assembly:

- **By position or search term:** Use the "position or search term" box to find areas of the genome associated with many different attributes, such as a specific chromosomal coordinate range; mRNA, EST, or STS marker names; or keywords from the GenBank description of an mRNA. [More information](#), including sample queries.
- **By gene name:** Type a gene name into the "search term" box, choose your gene from the drop-down list, then press "submit" to go directly to the assembly location associated with that gene. [More information](#).
- **By track type:** Click the "track search" button to find Genome Browser tracks that match specific selection criteria. [More information](#).

Download sequence and annotation data:

- **Using rsync** (recommended)
- **Using FTP**
- **Using UTTD**

基因组序列及注释信息下载

Part 1: 参考基因组版本及其来源的信息

Human Genome Browser - hg38 assembly

[view sequences](#)

UCSC Genome Browser assembly ID: hg38

Sequencing/Assembly provider ID: GRCh38 Genome Reference Consortium Human Reference 38 (GCA_000001405.15)

Assembly date: Dec. 2013

GenBank accession ID: GCA_000001305.2

NCBI Genome information: NCBI genome/51 (Homo sapiens)

NCBI Assembly information: NCBI assembly/883148 (GRCh38/GCA_000001405.15)

BioProject information: NCBI Bioproject: 31257



Part 2: 如何检索

Search the assembly:

Homo sapiens

(Graphic courtesy of CBSE)

- **By position or search term:** Use the "position or search term" box to find areas of the genome associated with many different attributes, such as a specific chromosomal coordinate range; mRNA, EST, or STS marker names; or keywords from the GenBank description of an mRNA. [More information](#), including sample queries.
- **By gene name:** Type a gene name into the "search term" box, choose your gene from the drop-down list, then press "submit" to go directly to the assembly location associated with that gene. [More information](#).
- **By track type:** Click the "track search" button to find Genome Browser tracks that match specific selection criteria. [More information](#).

Part 3: 如何下载

Download sequence and annotation data:

- [Using rsync](#) (recommended)
- [Using FTP](#)
- [Using HTTP](#)
- [Data use conditions and restrictions](#)
- [Acknowledgments](#)

内容提要

- UCSC Genome Browser 简介
- Genome Browser的使用：
 - 1) 页面布局介绍
 - 2) 如何有效检索
 - 3) 结果的显示和有效的调整

以人类基因组为例



Browse/Select Species

POPULAR SPECIES

Human Mouse Rat Fruitfly Worm Yeast

Enter species or common name

REPRESENTED SPECIES

Find Position

Human Assembly

Dec. 2013 (GRCh38/hg38)
Dec. 2013 (GRCh38/hg38)
Feb. 2009 (GRCh37/hg19)
Mar. 2006 (NCBI36/hg18)
May 2004 (NCBI35/hg17)
July 2003 (NCBI34/hg16)

GO

人类基因组拼接的版本号

Human Genome Browser - hg38 assembly

view sequences

UCSC Genome Browser assembly ID: hg38
Sequencing/Assembly provider ID: GRCh38 Genome Reference Consortium Human Reference 38 (GCA_000001405.15)
Assembly date: Dec. 2013
GenBank accession ID: GCA_000001305.2
NCBI Genome information: NCBI genome/51 (Homo sapiens)
NCBI Assembly information: NCBI assembly/883148 (GRCh38/GCA_000001405.15)
BioProject information: NCBI Bioproject: 31257

Homo sapiens

Step 1: 选定物种

Step 2: 选定基因组版本。这点非常重要！

在同一项研究项目中，对同一个物种，所采用的基因组版本应该是统一的。

Genome browser-检索词

- 1. 检索词可以表示基因位点

如chr2:207, 529, 892-207, 605, 560

- 2. 可以是基因名称

如CERB1

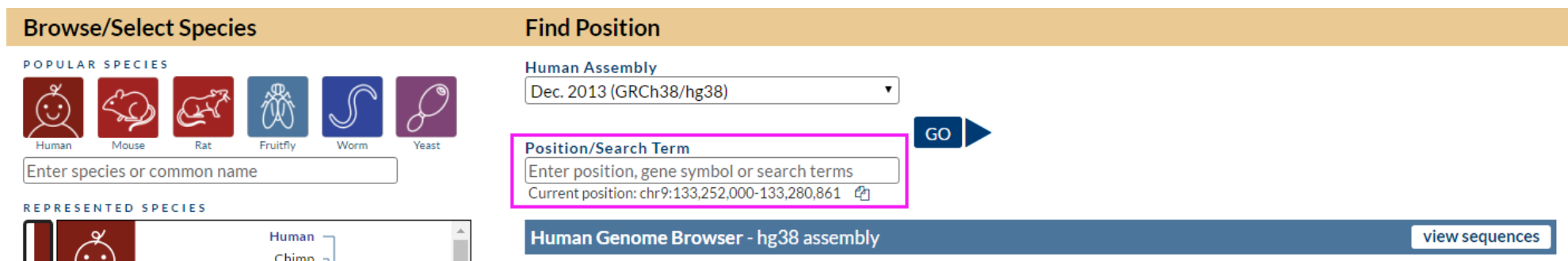
<http://genome.ucsc.edu/goldenPath/help/query.html>

- 3. 可以是GenBank ID

如AC008101

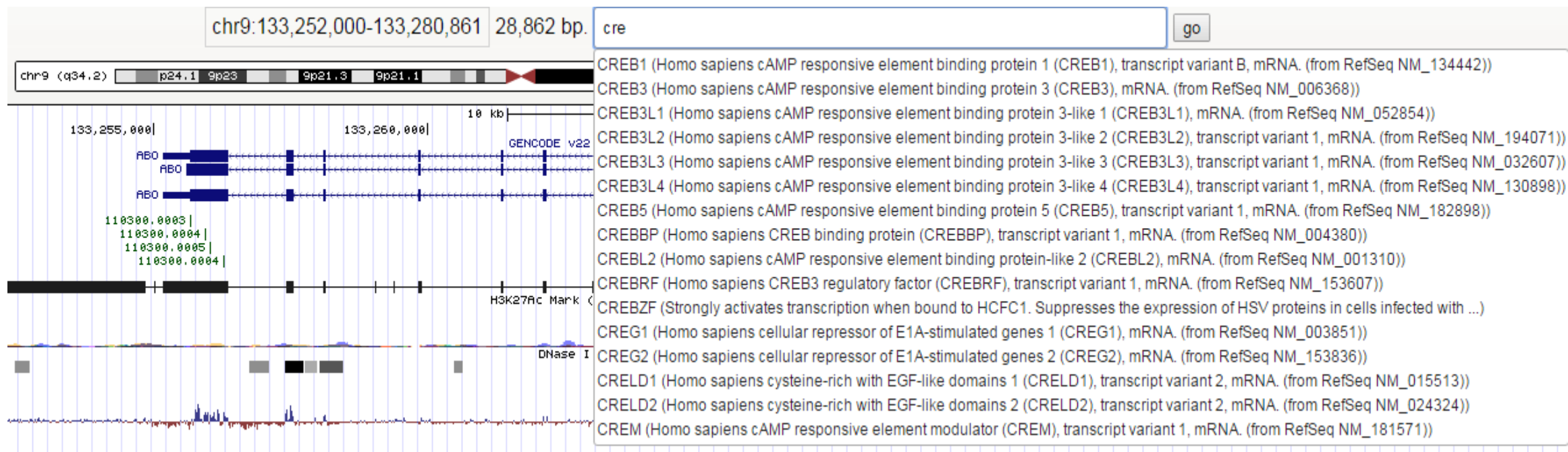
- 4. 可以是其他任意词汇

如HCC（原发性肝癌）



The screenshot displays the UCSC Genome Browser interface. It is divided into two main sections: 'Browse/Select Species' and 'Find Position'. The 'Browse/Select Species' section includes 'POPULAR SPECIES' with icons for Human, Mouse, Rat, Fruitfly, Worm, and Yeast, and a search bar 'Enter species or common name'. Below this is 'REPRESENTED SPECIES' showing a list with 'Human' and 'Chimp'. The 'Find Position' section features a dropdown menu for 'Human Assembly' set to 'Dec. 2013 (GRCh38/hg38)', a search box labeled 'Position/Search Term' with the placeholder 'Enter position, gene symbol or search terms' and the current position 'chr9:133,252,000-133,280,861', a 'GO' button, and a 'view sequences' link. The search box is highlighted with a pink border.

方法1: 键入部分基因名称, 系统会自动补全, 并显示下拉列表。
可在下拉列表中选择符合要求的词条。



注意：当键入的信息不完整时，搜索系统无法精确匹配，那么系统会自动跳转到可以部分匹配的列表里。我们可以在列表里进行进一步选择。

Warning/Error(s):

- Search terms are not very specific, only showing first 500 matching UCSC Genes.

OK

Known Genes

[CREB3L1 \(uc021qik.3\) at chr11:46277661-46321422](#) - Homo sapiens cAMP responsive element binding protein 3-like 1 (CREB3L1), mRNA. (from RefSeq NM_052854)

[CREB3 \(uc003zxv.4\) at chr9:35732335-35737004](#) - Homo sapiens cAMP responsive element binding protein 3 (CREB3), mRNA. (from RefSeq NM_006368)

[CREB3L2 \(uc003vtw.4\) at chr7:137874979-138002057](#) - Homo sapiens cAMP responsive element binding protein 3-like 2 (CREB3L2), transcript variant 1, mRNA. (from RefSeq NM_194071)

[CREB5 \(uc003szq.3\) at chr7:28412525-28825894](#) - Homo sapiens cAMP responsive element binding protein 5 (CREB5), transcript variant 1, mRNA. (from RefSeq NM_182898)

[CREB1 \(uc002vcc.4\) at chr2:207529892-207603431](#) - Homo sapiens cAMP responsive element binding protein 1 (CREB1), transcript variant B, mRNA. (from RefSeq NM_134442)

[CREB3L3 \(uc002lzl.4\) at chr19:4153601-4173054](#) - Homo sapiens cAMP responsive element binding protein 3-like 3 (CREB3L3), transcript variant 1, mRNA. (from RefSeq NM_032607)

[CREB3L4 \(uc001fdn.5\) at chr1:153967903-153974363](#) - Homo sapiens cAMP responsive element binding protein 3-like 4 (CREB3L4), transcript variant 1, mRNA. (from RefSeq NM_130898)

[CREB3 \(uc064syq.1\) at chr9:35732598-35736860](#) - cAMP responsive element binding protein 3 (from HGNC CREB3)

[CREB3 \(uc064syp.1\) at chr9:35732320-35737007](#) - The sequence shown here is derived from an Ensembl automatic analysis pipeline and should be considered as preliminary data. (from UniProt A0A087X1M)

[CREB3L2 \(uc064ijs.1\) at chr7:137999659-138002047](#) - cAMP responsive element binding protein 3-like 2 (from HGNC CREB3L2)

[CREB3L2 \(uc064ijr.1\) at chr7:137912834-137928269](#) - The sequence shown here is derived from an Ensembl automatic analysis pipeline and should be considered as preliminary data. (from UniProt H7C1W)

[CREB3L2 \(uc064ijq.1\) at chr7:137912824-137933890](#) - The sequence shown here is derived from an Ensembl automatic analysis pipeline and should be considered as preliminary data. (from UniProt C9JPH)

[CREB3L2 \(uc064ijp.1\) at chr7:137878841-138001866](#) - The sequence shown here is derived from an Ensembl automatic analysis pipeline and should be considered as preliminary data. (from UniProt A0A08)

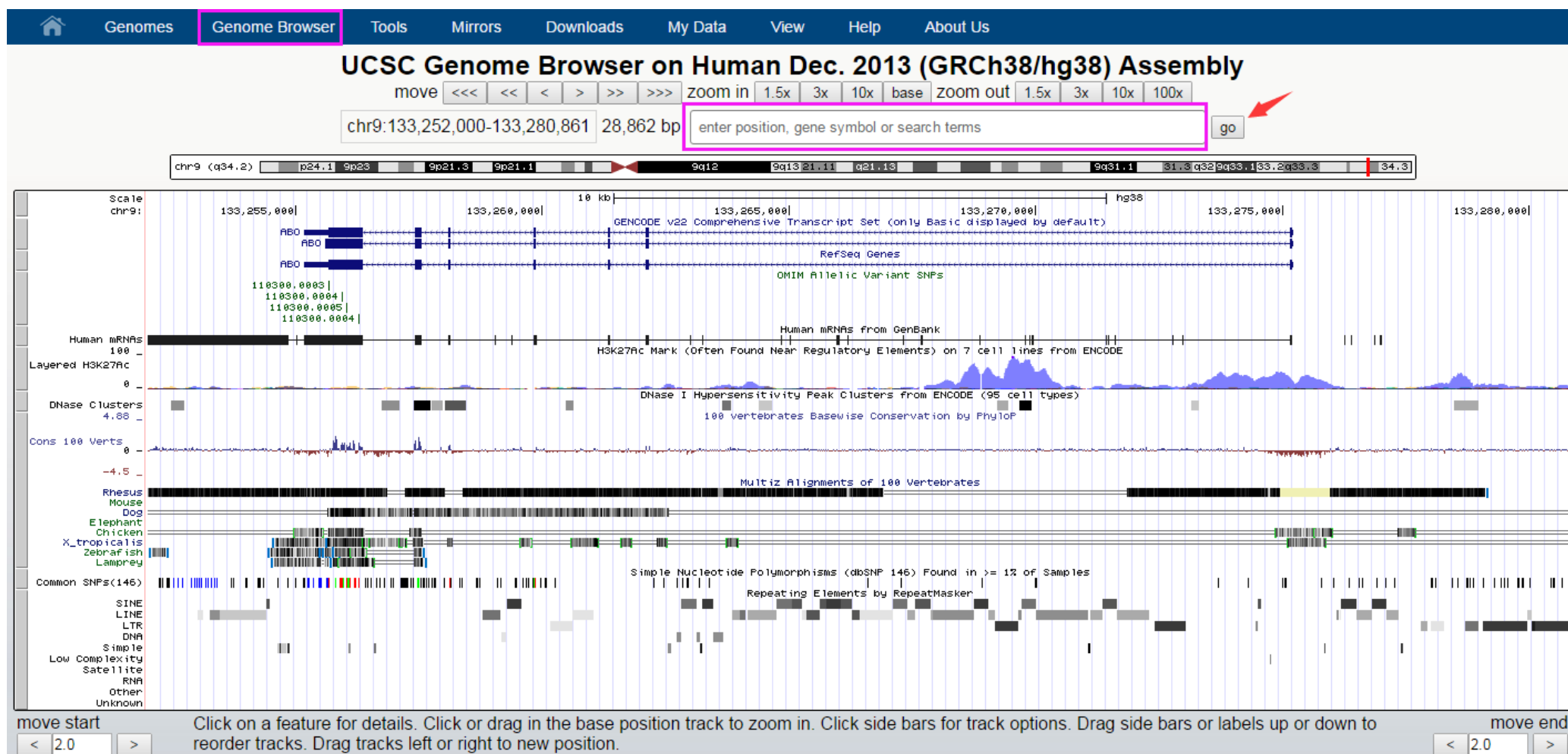
[CREB3L2 \(uc064ijo.1\) at chr7:137874981-138001827](#) - The sequence shown here is derived from an Ensembl automatic analysis pipeline and should be considered as preliminary data. (from UniProt A0A08)

[CREB5 \(uc064cjm.1\) at chr7:28809409-28819559](#) - The sequence shown here is derived from an Ensembl automatic analysis pipeline and should be considered as preliminary data. (from UniProt H0YFV2)

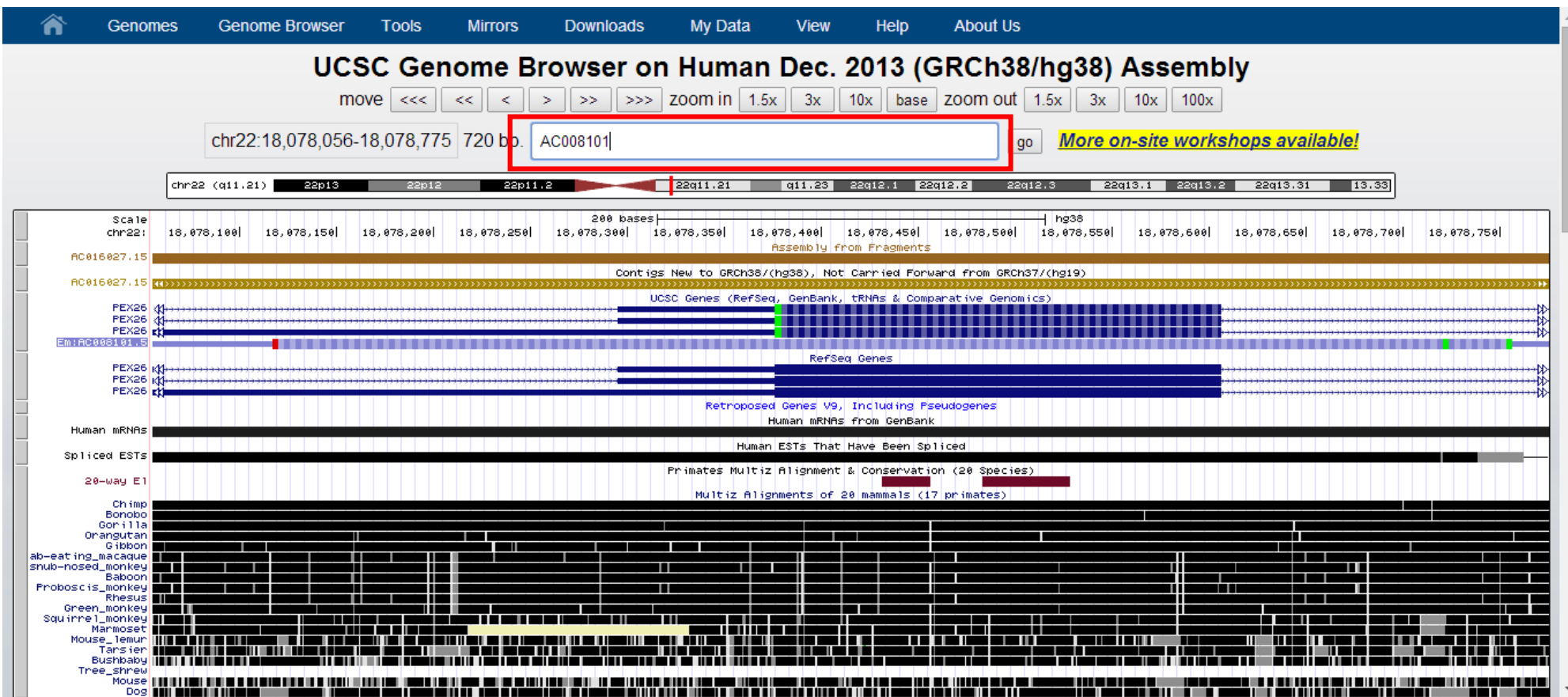
[CREB5 \(uc064cjm.1\) at chr7:28724300-28726731](#) - cAMP responsive element binding protein 5 (from HGNC CREB5)

[CREB5 \(uc064cjl.1\) at chr7:28686137-28779228](#) - cAMP responsive element binding protein 5 (from HGNC CREB5)

方法二：键入基因组的具体位置信息。
格式为：chr:start-end



方法三：键入 Accession number



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 - 3) **结果的显示和有效的调整**

Helix 解螺旋
Life



显示结果
调节窗口

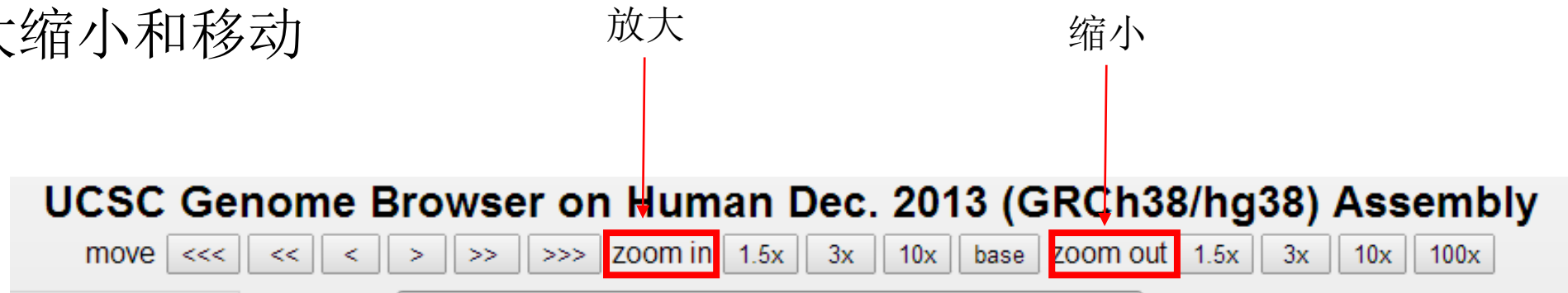
结果显示界面

- There are five main features on this page:
 - 1) a set of navigation controls,
 - 2) a chromosome ideogram,
 - 3) the annotations tracks image,
 - 4) display configuration buttons,
 - 5) a set of track display controls.

显示结果的调节

Feature 1: a set of navigation controls

放大缩小和移动

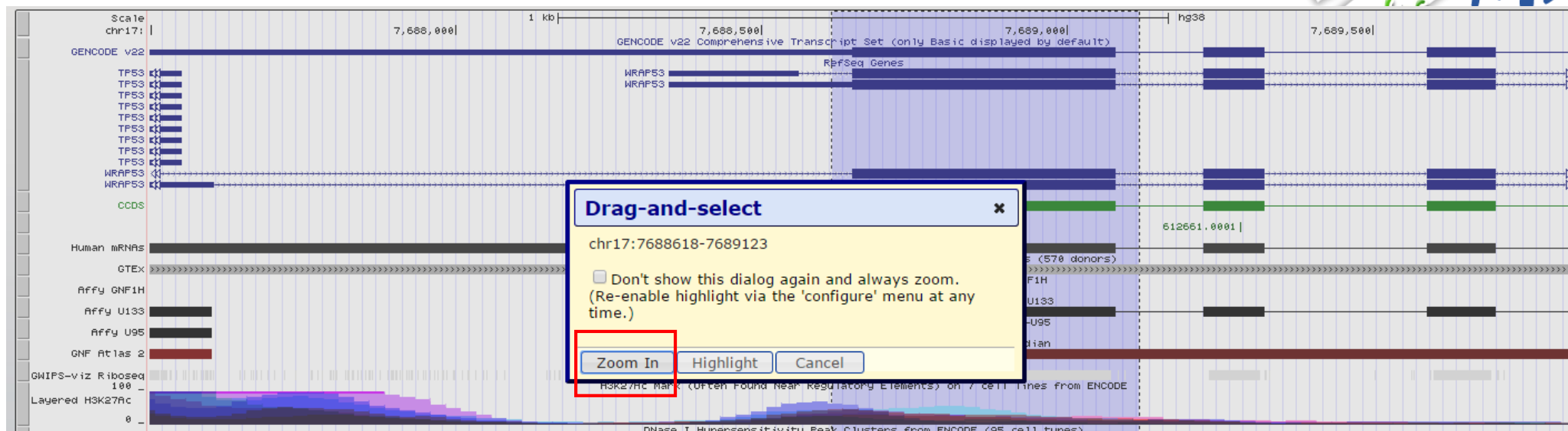


也可以直接在窗口里拖动。

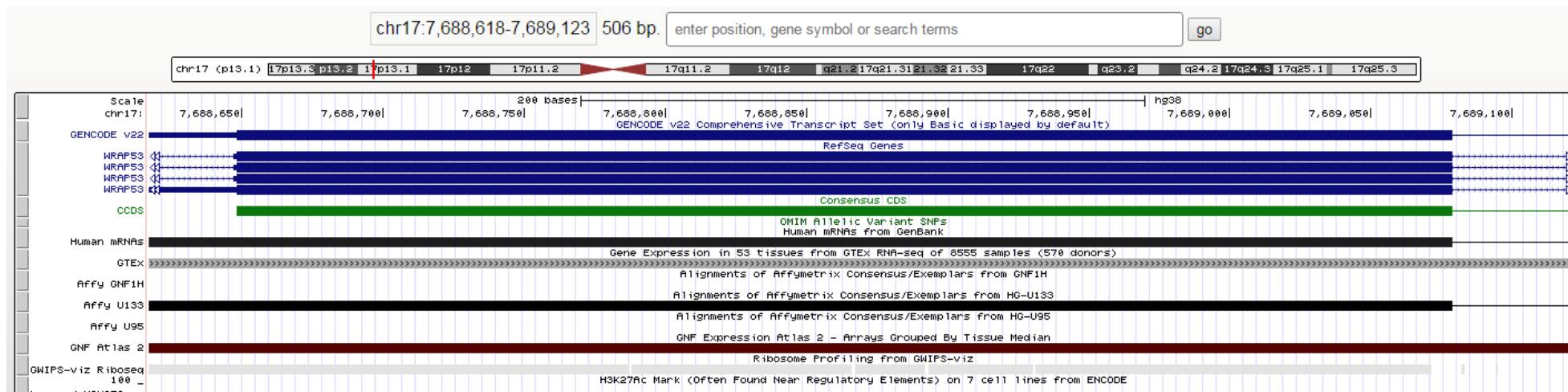
注意：根据选择显示的track信息和网速不同，直接拖动不一定很灵敏。

- 通过鼠标左键点击，选中感兴趣的区域来放大。如下图蓝色区域显示

Helix 解螺旋



放大蓝色选中区域



- 業施



UCSC Genome Browser on Human Dec. 2013 (GRCh38/hg38) Assembly

move <<< << < > >> >>> zoom in 1.5x 3x 10x base zoom out 1.5x 3x 10x 100x

chr17:7,672,893-7,685,854 12,962 bp. enter position, gene symbol or search terms go

chr17 (p13.1) 7p13.3 p13.2 1p13.1 17p12 17p11.2 17a11.2 17q12 q21.31 q21.32 q21.33 17q22 q23.2 q24.2 17q24.3 17q25.1 17q25.3

Scale chr17: 7,674,000 7,675,000 7,676,000 7,677,000 5 kb hg38

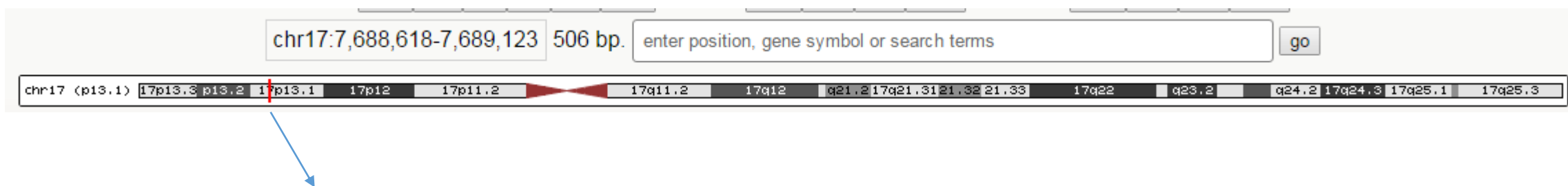
GENCODE v22 Comprehensive Transcript Set (only Basic displayed by default)

RP11-199F11.2

RefSeq Genes

显示结果的调节

Feature 2: a chromosome ideogram



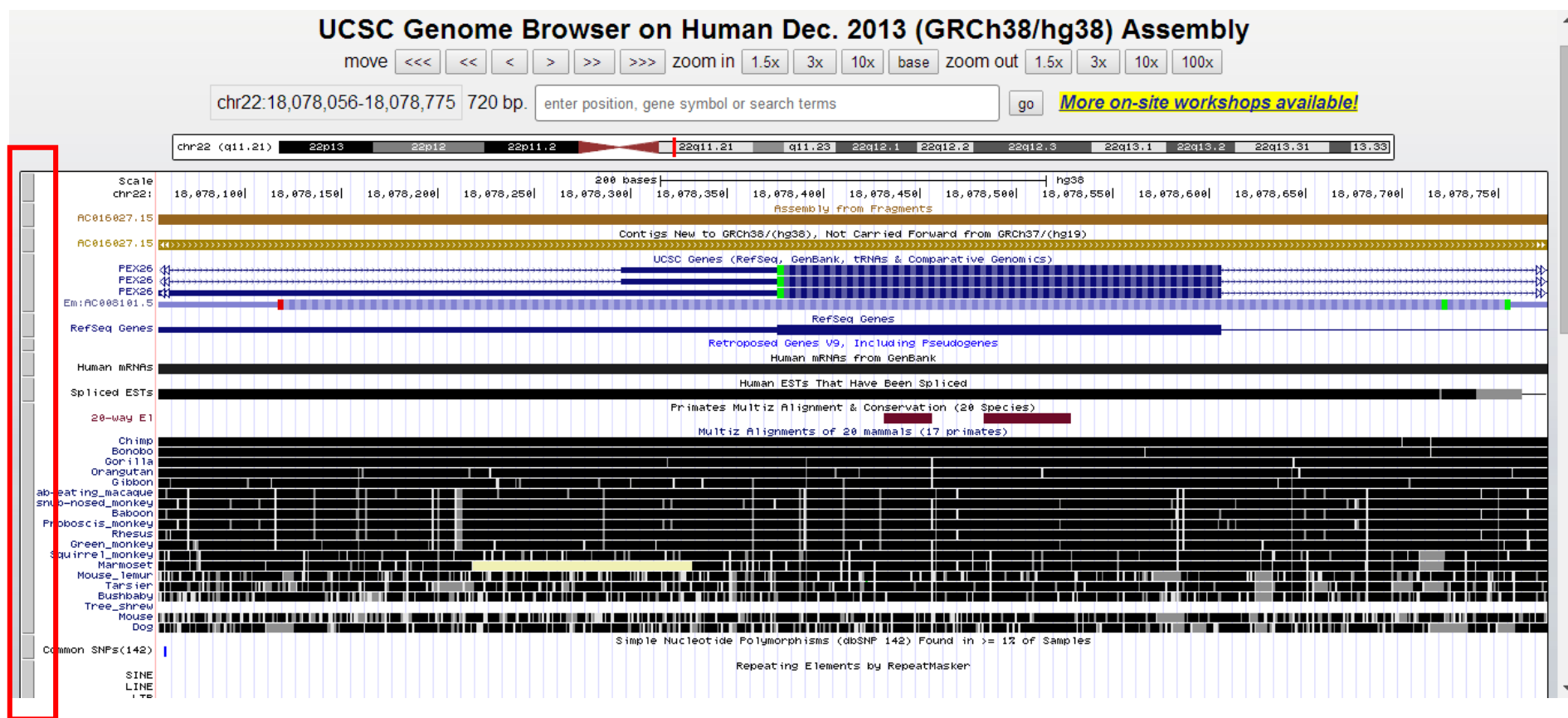
红线所标注的位置就是当前窗口在染色体上对应的具体位置

也可以直接在该染色体示意图上点击，移动到对应位置。

显示结果的调节



Feature3: the annotations tracks image (UCSC 特色)

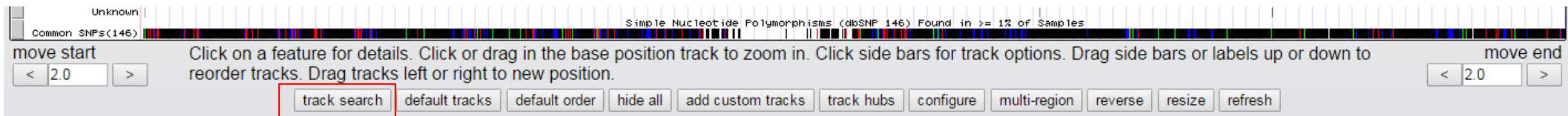


annotation tracks

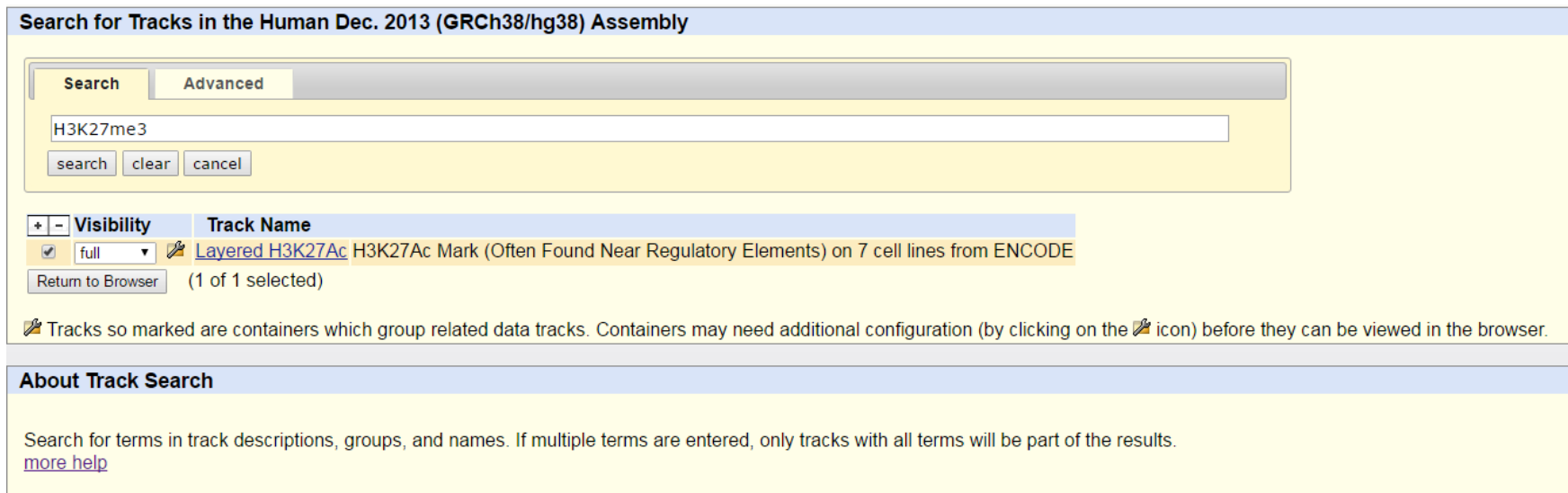
点击每条track和右侧方块，也可以调节结果显示

显示结果的调节

Feature 4: display configuration buttons



根据关键词查找特定 track



Configure Image

submit

image width: pixels

label area width: characters

text size:

- ☒ Display chromosome ideogram above main graphic
- ☒ Show light blue vertical guidelines, or light red vertical window separators in multi-region view
- ☒ Display labels to the left of items in tracks
- ☒ Display description above each track
- ☒ Show track controls under main graphic
- ☐ Next/previous item navigation
- ☒ Next/previous exon navigation
- ☒ Show exon numbers
- ☒ Enable highlight with drag-and-select (if unchecked, drag-and-select always zooms to selection)

Configure Tracks on UCSC Genome Browser: Human Dec. 2013 (GRCh38/hg38)

Tracks: Groups:

Control track and group visibility more selectively below.

- ☒ **Mapping and Sequencing**
- ☒ **Genes and Gene Predictions**
- ☒ **Phenotype and Literature**
- ☒ **mRNA and EST**
- ☒ **Expression**
- ☒ **Regulation**
- ☒ **Comparative Genomics**

Click to zoom in. Click side bars for track options. Drag side bars or labels up or

below and press refresh to alter tracks displayed.

Tracks will automatically be displayed in more compact modes.

通过统一的一个页面来调节各个track的显示方式。

显示结果的调节

Feature 4: display configuration buttons



- **Show exons using** -- display only the exonic regions
- **Show genes using** -- display only those regions with gene annotations in the specified genes track
- **Custom regions from BED URL** -- display only those regions listed in a specified BED file
- **Show one alternate haplotype** -- display the specified alternate haplotype in place on the reference chromosome (human assemblies hg17 and later only)

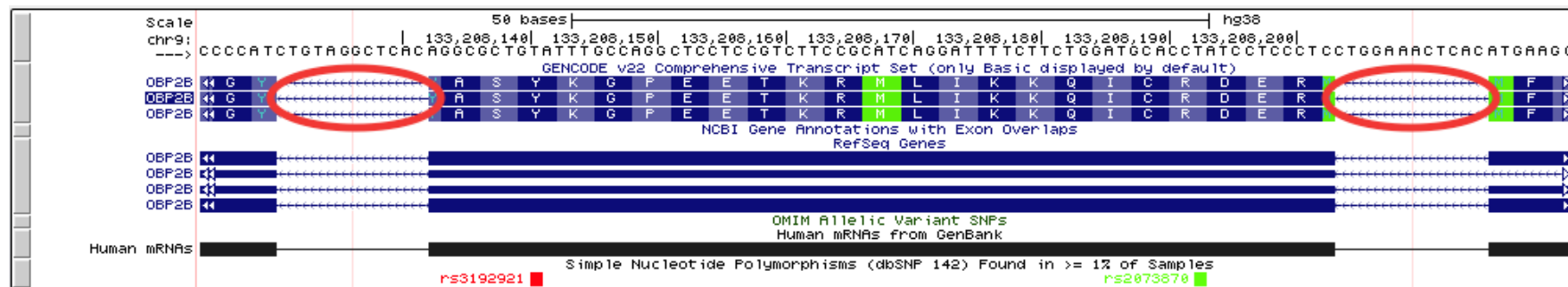
Configure Multi-Region View

☒ Exit multi-region mode
 ☐ Show exons using GENCODE v22. Use padding of: bases.
 ☐ Show genes using GENCODE v22. Use padding of: bases.
 ☐ Custom regions from BED URL:
☐ Show one alternate haplotype, placed on its chromosome, using ID:

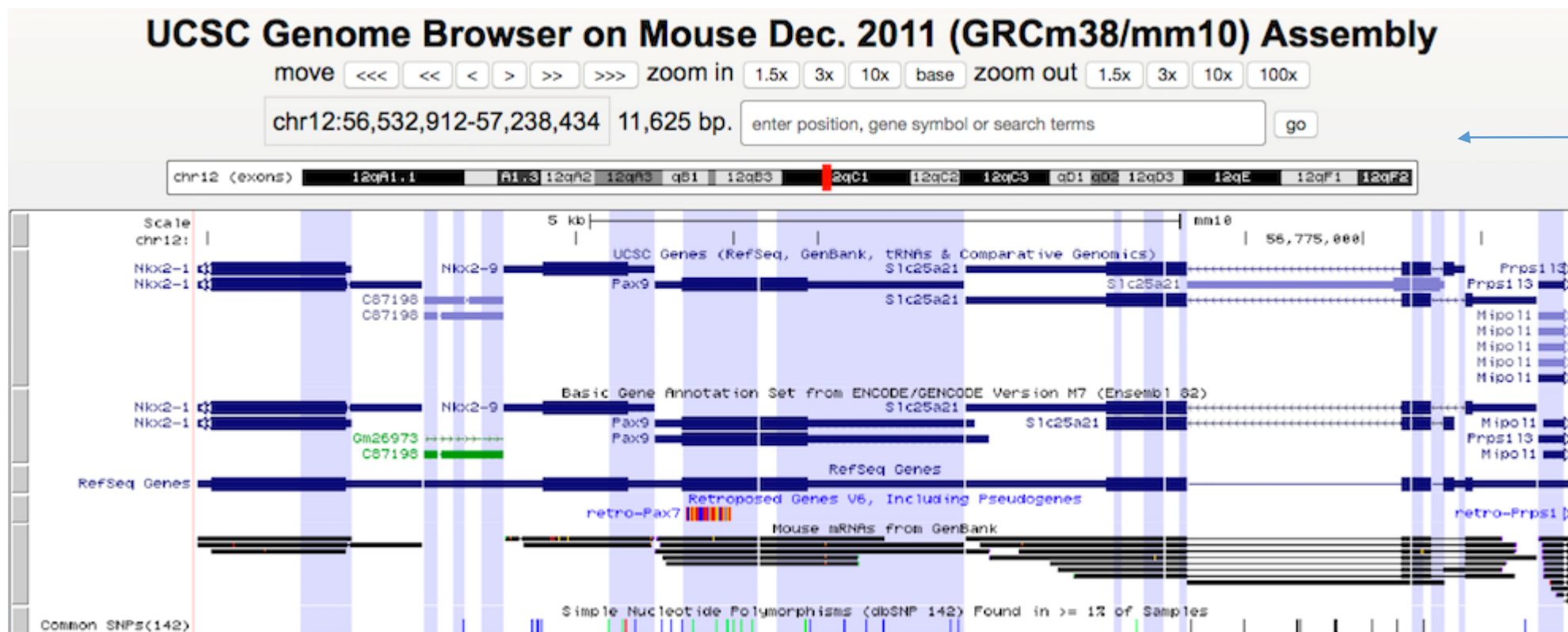
☐ Highlight alternating regions in multi-region view

[Help](#)

- Helix 解螺旋



红色的intron部分被去掉。



显示结果的调节

Feature 5: a set of track display controls

☐	Mapping and Sequencing	refresh
☐	Genes and Gene Predictions	refresh
☐	Phenotype and Literature	refresh
☐	mRNA and EST	refresh
☐	Expression	refresh
☐	Regulation	refresh
☐	Comparative Genomics	refresh
☐	Variation	refresh
☐	Repeats	refresh

Annotation Tracks的大类设置

Mapping and Sequencing

refresh

Base Position dense ▼	Assembly hide ▼	Hg19 Diff hide ▼	Alt Map... hide ▼	Centromeres hide ▼	Chromosome Band hide ▼
Clone Ends hide ▼	18 FISH Clones hide ▼	Gap hide ▼	GC Percent hide ▼	GRC Contigs hide ▼	GRC Incident hide ▼
GRC Patch Release [No data-chr17]	INSDC hide ▼	LRG Regions hide ▼	Restr Enzymes hide ▼	Scaffolds hide ▼	Short Match hide ▼
STS Markers hide ▼					

Genes and Gene Predictions

refresh

GENCODE v22 pack ▼	RefSeq Genes pack ▼	RetroGenes V9 hide ▼	All GENCODE... hide ▼	Augustus hide ▼	CCDS dense ▼
Geneid Genes hide ▼	Genscan Genes hide ▼	19 IKMC Genes Mapped hide ▼	LRG Transcripts hide ▼	MGC Genes hide ▼	Non-coding RNA... hide ▼
Old UCSC Genes hide ▼	ORFeome Clones hide ▼	Other RefSeq hide ▼	Pfam in UCSC Gene hide ▼	SGP Genes hide ▼	SIB Genes hide ▼
UCSC Alt Events hide ▼					

Phenotype and Literature

refresh

OMIM AV SNPs pack ▼	ClinGen CNVs hide ▼	ClinVar Variants hide ▼	10 Coriell CNVs hide ▼	Development Delay hide ▼	GeneReviews dense ▼
GWAS Catalog hide ▼	OMIM Genes hide ▼	OMIM Pheno Loci hide ▼			

mRNA and EST

refresh

Human mRNAs dense ▼	Human ESTs hide ▼	Other ESTs hide ▼	Other mRNAs hide ▼	SIB Alt-Splicing hide ▼	Spliced ESTs hide ▼
--	--------------------------------------	--------------------------------------	---------------------------------------	--	--

Expression

refresh

GTEx dense ▼	Affy GNF1H dense ▼	Affy U133 dense ▼	Affy U95 dense ▼	GNF Atlas 2 dense ▼	GWIPS-viz Riboseq dense ▼
---------------------------------	---------------------------------------	--------------------------------------	-------------------------------------	--	--

Regulation

refresh

★ ENCODE Regulation... show ▼	CpG Islands... hide ▼
--	--

Comparative Genomics

refresh

Conservation full ▼	Cons 20 Mammals hide ▼	Cons 7 Verts hide ▼	Primate Chain/Net hide ▼	Placental Chain/Net hide ▼	Vertebrate Chain/Net hide ▼
--	---	--	---	---	--

—
Variation
refresh

Common SNPs(146) pack ▼	All SNPs(141) hide ▼	All SNPs(142) hide ▼	All SNPs(144) hide ▼	All SNPs(146) hide ▼	Common SNPs(141) hide ▼
Common SNPs(142) hide ▼	Common SNPs(144) hide ▼	DGV Struct Var hide ▼	Flagged SNPs(141) hide ▼	Flagged SNPs(142) hide ▼	Flagged SNPs(144) hide ▼
Flagged SNPs(146) hide ▼	Mult. SNPs(141) hide ▼	Mult. SNPs(142) hide ▼	Mult. SNPs(144) hide ▼	Mult. SNPs(146) hide ▼	

—
Repeats
refresh

RepeatMasker full ▼	RepeatMasker Viz. hide ▼	Interrupted Rpts hide ▼	Microsatellite hide ▼	Segmental Dups hide ▼	Self Chain hide ▼
Simple Repeats hide ▼	WM + SDust hide ▼				



- 这些都是来自世界各地的数据库的注释信息，大家可以根据自己的需要进行进一步的筛选。
- 具体的信息可以点击链接进入详细的介绍页面。每一个注释集都有自己独立的介绍页面。



GeneReviews Track Settings

GeneReviews (▲[All Phenotype and Literature tracks](#))

Display mode:

[View table schema](#)

Data last updated: 2016-06-07

Description

[GeneReviews](#) is an online collection of expert-authored, peer-reviewed articles that describe specific gene-related diseases. *GeneReviews* articles are searchable by disease name, gene symbol, protein name, author, or title. *GeneReviews* is supported by the National Institutes of Health, hosted at NCBI as part of the [Genetic Testing Registry \(GTR\)](#). The *GeneReviews* data underlying this track will be updated frequently.

The GeneReviews track allows the user to locate the NCBI *GeneReviews* resource quickly from the Genome Browser. A condensed version of the *GeneReviews* article name and its related diseases are displayed on the item details page as links. Similar information, when available, is provided in the details page of items from the UCSC Genes, RefSeq Genes, and OMIM Genes tracks.

References

Pagon RA, Adam MP, Bird TD, *et al.*, editors. GeneReviews® [Internet]. Seattle (WA): University of Washington, Seattle; 1993-2014. Available from: <http://www.ncbi.nlm.nih.gov/books/NBK1116>.

GeneReviews (BRCA1)

Item: BRCA1

Position: [chr17:43044295-43125370](#)

Band: 17q21.31

Genomic Size: 81076

[View DNA for this feature](#) (hg38/Human)

GeneReviews available for BRCA1: (Click GR short name)

GR short name

Disease name

[brca1](#)

BRCA1 and BRCA2 Hereditary Breast/Ovarian Cancer

[View table schema](#)

[Go to GeneReviews track controls](#)

Data last updated: 2016-06-14



GeneReviews® [Internet].

[Show details](#)

[GeneReviews by Title](#) (v)

[GeneReviews Advanced Search](#) [Help](#)

Search GeneReviews

< Prev

Next >



BRCA1 and BRCA2 Hereditary Breast and Ovarian Cancer

Synonym: HBOC

Nancie Petrucelli, MS, Mary B Daly, MD, PhD, and Gerald L Feldman, MD, PhD, FACMG.

[Author Information](#)

Initial Posting: September 4, 1998; Last Update: September 26, 2013.

Summary

Go to: (v)

Clinical characteristics. Hereditary breast and ovarian cancer syndrome (HBOC), caused by a [germline](#) pathogenic variant in *BRCA1* or *BRCA2*, is characterized by an increased risk for breast cancer, ovarian cancer, prostate cancer, and pancreatic cancer. The lifetime risk for these cancers in individuals with a pathogenic variant in *BRCA1* or *BRCA2*:

- 40%-80% for breast cancer
- 11%-40% for ovarian cancer
- 1%-10% for male breast cancer
- Up to 39% for prostate cancer
- 1%-7% for pancreatic cancer

Individuals with *BRCA2* pathogenic variants may also be at an increased risk for melanoma. Prognosis for *BRCA1/2*-related cancer depends on the stage at which the cancer is diagnosed; however, studies on survival have revealed conflicting results for individuals with [germline](#) *BRCA1* or *BRCA2* pathogenic variants when compared to controls.

Diagnosis/testing. An increased likelihood of a *BRCA1* or *BRCA2* pathogenic variant is suspected on the basis of certain personal and [family history](#) characteristics and various clinical criteria. A diagnosis of HBOC is made following [molecular genetic testing](#) in an individual or family with a [germline](#) *BRCA1* or *BRCA2* pathogenic variant. No currently available technique can guarantee the identification of all cancer-predisposing allelic variants in *BRCA1* or *BRCA2*.

设置显示方式

+

Mapping and Sequencing

refresh

-

Genes and Gene Predictions

refresh

UCSC Genes dense ▼ hide dense squish pack full	RefSeq Genes dense ▼	RetroGenes V9 pack ▼	CCDS hide ▼	GENCODE (Ensembl)... hide ▼	Genscan Genes hide ▼
Transcripts hide ▼	MGC Genes hide ▼	ORFeome Clones hide ▼	Other RefSeq hide ▼	Pfam in UCSC Gene hide ▼	SIB Genes hide ▼
UCSC Alt Events hide ▼					

full

hide

dense

squish

pack

full

Hide: 不显示

Base Position: 38460000 38470000 38480000 38490000 38500000 38510000 38520000 38530000

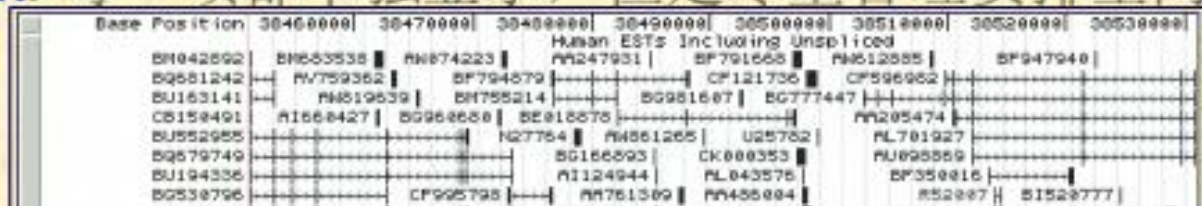
Dense: 所有信息在一条直线里显示出来

Base Position: 38460000 38470000 38480000 38490000 38500000 38510000 38520000 38530000
Human ESTs including Unspliced

Full: 每一项都用一条直线显示



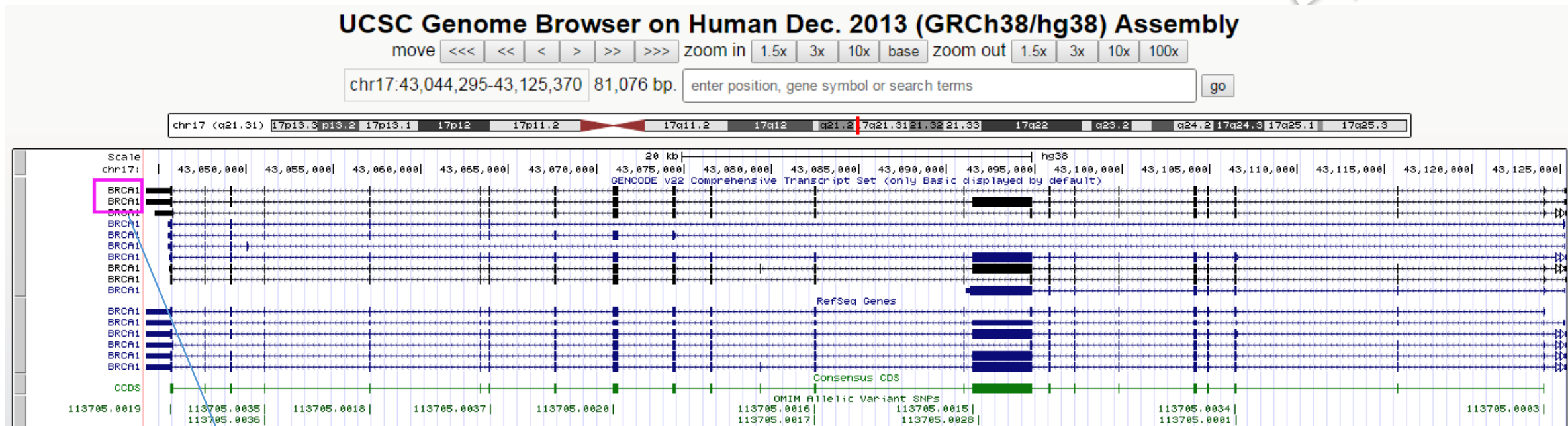
Pack: 每一项都单独显示, 但是尽量合理安排空间



Squish: 和pack类似, 但是高度只有一半



基因相关信息的大整合



单击基因名称的track，会进入到基因信息的详细整合页面

Human Gene BRCA1 (ENST00000352993.6) Description and Page Index

Description: E3 ubiquitin-protein ligase that specifically mediates the formation of 'Lys-6'-linked polyubiquitin chains and plays a central role in DNA repair by facilitating cellular responses to DNA damage. It is unclear whether it also mediates the formation of other types of polyubiquitin chains. The E3 ubiquitin-protein ligase activity is required for its tumor suppressor function. The BRCA1-BARD1 heterodimer coordinates a diverse range of cellular pathways such as DNA damage repair, ubiquitination and transcriptional regulation to maintain genomic stability. Regulates centrosomal microtubule nucleation. Required for normal cell cycle progression from G2 to mitosis. Required for appropriate cell cycle arrests after ionizing irradiation in both the S-phase and the G2 phase of the cell cycle. Involved in transcriptional regulation of P21 in response to DNA damage. Required for FANCD2 targeting to sites of DNA damage. May function as a transcriptional regulator. Inhibits lipid synthesis by binding to inactive phosphorylated ACACA and preventing its dephosphorylation. Contributes to homologous recombination repair (HRR) via its direct interaction with PALB2, fine-tunes recombinational repair partly through its modulatory role in the PALB2-dependent loading of BRCA2-RAD51 repair machinery at DNA breaks. Component of the BRCA1-RBBP8 complex which regulates CHEK1 activation and controls cell cycle G2/M checkpoints on DNA damage via BRCA1-mediated ubiquitination of RBBP8. (from UniProt P38398)

RefSeq Summary (NM_007294): This gene encodes a nuclear phosphoprotein that plays a role in maintaining genomic stability, and it also acts as a tumor suppressor. The encoded protein combines with other tumor suppressors, DNA damage sensors, and signal transducers to form a large multi-subunit protein complex known as the BRCA1-associated genome surveillance complex (BASC). This gene product associates with RNA polymerase II, and through the C-terminal domain, also interacts with histone deacetylase complexes. This protein thus plays a role in transcription, DNA repair of double-stranded breaks, and recombination. Mutations in this gene are responsible for approximately 40% of inherited breast cancers and more than 80% of inherited breast and ovarian cancers. Alternative splicing plays a role in modulating the subcellular localization and physiological function of this gene. Many alternatively spliced transcript variants, some of which are disease-associated mutations, have been described for this gene, but the full-length natures of only some of these variants has been described. A related pseudogene, which is also located on chromosome 17, has been identified. [provided by RefSeq, May 2009].

Gencode Transcript: ENST00000352993.6

Transcript (Including UTRs)

Position: hg38 chr17:43,044,295-43,125,370 **Size:** 81,076 **Total Exon Count:** 22 **Strand:** -

Coding Region

Position: hg38 chr17:43,045,678-43,124,096 **Size:** 78,419 **Coding Exon Count:** 21

基因功能描述

Page Index	Sequence and Links	UniProtKB Comments	MalaCards	CTD	RNA-Seq Expression
Microarray Expression	RNA Structure	Protein Structure	Other Species	GO Annotations	mRNA Descriptions
Pathways	Other Names	GeneReviews	Methods		

Data last updated: 2015-04-24

各类资源整合

通过UCSC Genome Browser， 我们能对想查找的基因进行非常全面的了解。

内容提要

- UCSC Genome Browser 简介
- Genome Browser的使用
- Table Browser 的使用
- Download genome annotation



Table Browser

- 提供了访问数据库的便利入口
- 用文本形式来获取储存在Genome Browser数据库中的基因组汇编和注释数据

Table Browser-功能

- 为整个染色体或者一些特定的序列获取DNA序列信息或者注释信息
- 用特定的条件对输出结果进行筛选
- 创建自己的路径，并在Genome Browser中显示出来

Table Browser-功能

- 整合多重查询，并为其产生相关的输出
- 为选定的数据集提供基本的统计结果
- 显示一个数据表格的详细情况，并且列出在数据库中与其相关的所有表格
- 根据其他应用程序和数据库的要求，对输出结果进行格式化

实例

- 下载人类第17号染色体上简单重复10次以上的序列

Clade:mammal



Table Browser

Use this program to retrieve the data associated with a track in text format, to calculate intersections between tracks, and to retrieve DNA sequence covered by a track. For help in using this application see [Using the Table Browser](#) for a description of the controls in this form, the [User's Guide](#) for general information and sample queries, and the OpenHelix Table Browser [tutorial](#) for a narrated presentation of the software features and usage. For more complex queries, you may want to use [Galaxy](#) or our [public MySQL server](#). To examine the biological function of your set through annotation enrichments, send the data to [GREAT](#). Send data to [GenomeSpace](#) for use with diverse computational tools. Refer to the [Credits](#) page for the list of contributors and usage restrictions associated with these data. All tables can be downloaded in their entirety from the [Sequence and Annotation Downloads](#) page.

clade: **genome:** **assembly:**

group: **track:**

table:

region:

identifiers (sessions):

filter:

intersection:

correlation:

output format: Send output to ☐ [Galaxy](#) ☐ [GREAT](#) ☐ [GenomeSpace](#)

output file: (leave blank to keep output in browser)

file type returned: ☒ plain text ☐ gzip compressed

To reset **all** user cart settings (including custom tracks), [click here](#).

Genome:human

Table Browser

Use this program to retrieve the data associated with a track in text format, to calculate intersections between tracks, and to retrieve DNA sequence covered by a track. For help in using this application see [Using the Table Browser](#) for a description of the controls in this form, the [User's Guide](#) for general information and sample queries, and the OpenHelix Table Browser [tutorial](#) for a narrated presentation of the software features and usage. For more complex queries, you may want to use [Galaxy](#) or our [public MySQL server](#). To examine the biological function of your set through annotation enrichments, send the data to [GREAT](#). Send data to [GenomeSpace](#) for use with diverse computational tools. Refer to the [Credits](#) page for the list of contributors and usage restrictions associated with these data. All tables can be downloaded in their entirety from the [Sequence and Annotation Downloads](#) page.

clade: Mammal

genome: Human

assembly: Dec. 2013 (GRCh38/hg38)

group: Repeats

table: simpleRepeat describe table

region: ☐ genome ☒ position chr

identifiers (names/accessions):

filter: edit clear

intersection: create

correlation: create

output format: all fields from selected

output file:

file type returned: ☒ plain text ☐ Hedgehog

get output
summary/statistics

Human

Mouse

Alpaca

Armadillo

Baboon

Bushbaby

Cat

Chimp

Chinese hamster

Cow

Dog

Dolphin

Elephant

Ferret

Gibbon

Gorilla

Guinea pig

Hedgehog

Horse

Kangaroo rat

peats

add custom tracks

track hubs

lookup

define regions

Send output to ☐ [Galaxy](#) ☐ [GREAT](#) ☐ [GenomeSpace](#)

☐
to keep output in browser)

To reset all user cart settings (including custom tracks), [click here](#).

Assembly:hg38



Table Browser

Use this program to retrieve the data associated with a track in text format, to calculate intersections between tracks, and to retrieve DNA sequence covered by a track. For help in using this application see [Using the Table Browser](#) for a description of the controls in this form, the [User's Guide](#) for general information and sample queries, and the OpenHelix Table Browser [tutorial](#) for a narrated presentation of the software features and usage. For more complex queries, you may want to use [Galaxy](#) or our [public MySQL server](#). To examine the biological function of your set through annotation enrichments, send the data to [GREAT](#). Send data to [GenomeSpace](#) for use with diverse computational tools. Refer to the [Credits](#) page for the list of contributors and usage restrictions associated with these data. All tables can be downloaded in their entirety from the [Sequence and Annotation Downloads](#) page.

clade: **genome:** **assembly:**

group: **track:**

table:

region: ☐ genome ☒ position

identifiers (names/accessions):

filter:

intersection:

correlation:

output format: Send output to ☐ [Galaxy](#) ☐ [GREAT](#) ☐ [GenomeSpace](#)

output file: (leave blank to keep output in browser)

file type returned: ☒ plain text ☐ gzip compressed

To reset **all** user cart settings (including custom tracks), [click here](#).

Group:repeats

Table Browser

Use this program to retrieve the data associated with a track in text format, to calculate intersections between tracks, and to retrieve DNA sequence covered by a track. For help in using this application see [Using the Table Browser](#) for a description of the controls in this form, the [User's Guide](#) for general information and sample queries, and the OpenHelix Table Browser [tutorial](#) for a narrated presentation of the software features and usage. For more complex queries, you may want to use [Galaxy](#) or our [public MySQL server](#). To examine the biological function of your set through annotation enrichments, send the data to [GREAT](#). Send data to [GenomeSpace](#) for use with diverse computational tools. Refer to the [Credits](#) page for the list of contributors and usage restrictions associated with these data. All tables can be downloaded in their entirety from the [Sequence and Annotation Downloads](#) page.

clade: Mammal **genome:** Human **assembly:** Dec. 2013 (GRCh38/hg38)

group: Repeats **track:** Simple Repeats add custom tracks track hubs

table: Mapping and Sequencing Genes and Gene Predictions Phenotype and Literature mRNA and EST Expression Regulation Comparative Genomics Variation Repeats All Tracks All Tables

region: 17:1-83257441 lookup define regions

identify: paste list upload list

filter: Comparative Genomics Variation Repeats All Tracks All Tables

intersection: Repeats All Tracks All Tables

correlation: All Tables

output format: all fields from selected table Send output to ☐ [Galaxy](#) ☐ [GREAT](#) ☐ [GenomeSpace](#)

output file: (leave blank to keep output in browser)

file type returned: ☒ plain text ☐ gzip compressed

get output summary/statistics

To reset **all** user cart settings (including custom tracks), [click here](#).

Track: simple repeats

Table Browser

Use this program to retrieve the data associated with a track in text format, to calculate intersections between tracks, and to retrieve DNA sequence covered by a track. For help in using this application see [Using the Table Browser](#) for a description of the controls in this form, the [User's Guide](#) for general information and sample queries, and the OpenHelix Table Browser [tutorial](#) for a narrated presentation of the software features and usage. For more complex queries, you may want to use [Galaxy](#) or our [public MySQL server](#). To examine the biological function of your set through annotation enrichments, send the data to [GREAT](#). Send data to [GenomeSpace](#) for use with diverse computational tools. Refer to the [Credits](#) page for the list of contributors and usage restrictions associated with these data. All tables can be downloaded in their entirety from the [Sequence and Annotation Downloads](#) page.

clade: Mammal **genome:** Human **assembly:** Dec. 2013 (GRCh38/hg38)

group: Repeats **track:** Simple Repeats add custom tracks track hubs

table: simpleRepeat describe table schema

region: genome position chr17:1-832574 define regions

identifiers (names/accessions): paste list

filter: edit clear

intersection: create

correlation: create

output format: all fields from selected table Send output to ☐ Galaxy ☐ GREAT ☐ GenomeSpace

output file: (leave blank to keep output in browser)

file type returned: ☒ plain text ☐ gzip compressed

get output summary/statistics

To reset **all** user cart settings (including custom tracks), [click here](#).

Table for output

Table Browser

Use this program to retrieve the data associated with a track in text format, to calculate intersections between tracks, and to retrieve DNA sequence covered by a track. For help in using this application see [Using the Table Browser](#) for a description of the controls in this form, the [User's Guide](#) for general information and sample queries, and the OpenHelix Table Browser [tutorial](#) for a narrated presentation of the software features and usage. For more complex queries, you may want to use [Galaxy](#) or our [public MySQL server](#). To examine the biological function of your set through annotation enrichments, send the data to [GREAT](#). Send data to [GenomeSpace](#) for use with diverse computational tools. Refer to the [Credits](#) page for the list of contributors and usage restrictions associated with these data. All tables can be downloaded in their entirety from the [Sequence and Annotation Downloads](#) page.

clade: Mammal **genome:** Human **assembly:** Dec. 2013 (GRCh38/hg38)

group: Repeats **track:** Simple Repeats add custom tracks track hubs

table: simpleRepeat describe table schema

region: genome position chr17 lookup define regions

identifiers (names/accessions): paste list upload list

filter: edit clear

intersection: create

correlation: create

output format: all fields from selected table Send output to ☐ [Galaxy](#) ☐ [GREAT](#) ☐ [GenomeSpace](#)

output file: (leave blank to keep output in browser)

file type returned: ☒ plain text ☐ gzip compressed

get output summary/statistics

To reset **all** user cart settings (including custom tracks), [click here](#).

Table for simple repeats

Schema for Simple Repeats - Simple Tandem Repeats by TRF

Database: hg38 **Primary Table:** simpleRepeat **Row Count:** 1,014,212 **Data last updated:** 2013-12-29

Format description: Describes the Simple Tandem Repeats

field	example	SQL type	description
bin	585	smallint(5) unsigned	Indexing field to speed chromosome range queries.
chrom	chr1	varchar(255)	Reference sequence chromosome or scaffold
chromStart	10000	int(10) unsigned	Start position in chromosome
chromEnd	10468	int(10) unsigned	End position in chromosome
name	trf	varchar(255)	Simple Repeats tag name
period	6	int(10) unsigned	Length of repeat unit
copyNum	77.2	float	Mean number of copies of repeat
consensusSize	6	int(10) unsigned	Length of consensus sequence
perMatch	95	int(10) unsigned	Percentage Match
perIndel	3	int(10) unsigned	Percentage Indel
score	789	int(10) unsigned	Alignment Score = 2*match-7*mismatch-7*indel; minscore=50
A	33	int(10) unsigned	Percent of A's in repeat unit
C	51	int(10) unsigned	Percent of C's in repeat unit
G	0	int(10) unsigned	Percent of G's in repeat unit
T	15	int(10) unsigned	Percent of T's in repeat unit
entropy	1.43	float	Entropy
sequence	TAACCC	longblob	Sequence of repeat unit element

Position:chr17



Table Browser

Use this program to retrieve the data associated with a track in text format, to calculate intersections between tracks, and to retrieve DNA sequence covered by a track. For help in using this application see [Using the Table Browser](#) for a description of the controls in this form, the [User's Guide](#) for general information and sample queries, and the OpenHelix Table Browser [tutorial](#) for a narrated presentation of the software features and usage. For more complex queries, you may want to use [Galaxy](#) or our [public MySQL server](#). To examine the biological function of your set through annotation enrichments, send the data to [GREAT](#). Send data to [GenomeSpace](#) for use with diverse computational tools. Refer to the [Credits](#) page for the list of contributors and usage restrictions associated with these data. All tables can be downloaded in their entirety from the [Sequence and Annotation Downloads](#) page.

clade: **genome:** **assembly:**

group: **track:**

table:

region: ☐ genome ☒ position

identifiers (names/accessions):

filter:

intersection:

correlation:

output format: Send output to ☐ [Galaxy](#) ☐ [GREAT](#) ☐ [GenomeSpace](#)

output file: (leave blank to keep output in browser)

file type returned: ☒ plain text ☐ gzip compressed

To reset **all** user cart settings (including custom tracks), [click here](#).

点击filter:edit

Table Browser

Use this program to retrieve the data associated with a track in text format, to calculate intersections between tracks, and to retrieve DNA sequence covered by a track. For help in using this application see [Using the Table Browser](#) for a description of the controls in this form, the [User's Guide](#) for general information and sample queries, and the OpenHelix Table Browser [tutorial](#) for a narrated presentation of the software features and usage. For more complex queries, you may want to use [Galaxy](#) or our [public MySQL server](#). To examine the biological function of your set through annotation enrichments, send the data to [GREAT](#). Send data to [GenomeSpace](#) for use with diverse computational tools. Refer to the [Credits](#) page for the list of contributors and usage restrictions associated with these data. All tables can be downloaded in their entirety from the [Sequence and Annotation Downloads](#) page.

clade:
genome:
assembly:

group:
track:

table:

region: ☐ genome ☒ position

identifiers (names/accessions):

filter:

intersection:

correlation:

output format:
Send output to ☐ [Galaxy](#) ☐ [GREAT](#) ☐ [GenomeSpace](#)

output file: (leave blank to keep output in browser)

file type returned: ☒ plain text ☐ gzip compressed

To reset **all** user cart settings (including custom tracks), [click here](#).

筛选条件: copyNum>10

Filter on Fields from hg38.simpleRepeat

bin	is	ignore	0	
chrom	does	match	*	AND
chromStart	is	ignore	0	AND
chromEnd	is	ignore	0	AND
name	does	match	*	AND
period	is	ignore	0	AND
copyNum	is	>	10	AND
consensusSize	is	ignore	0	AND
perMatch	is	ignore	0	AND
perIndel	is	ignore	0	AND
score	is	ignore	0	AND
A	is	ignore	0	AND
C	is	ignore	0	AND
G	is	ignore	0	AND
T	is	ignore	0	AND
entropy	is	ignore	0	AND
sequence	does	match	*	

AND Free-form query:

submit cancel

File type: plain txt

Table Browser

Use this program to retrieve the data associated with a track in text format, to calculate intersections between tracks, and to retrieve DNA sequence covered by a track. For help in using this application see [Using the Table Browser](#) for a description of the controls in this form, the [User's Guide](#) for general information and sample queries, and the OpenHelix Table Browser [tutorial](#) for a narrated presentation of the software features and usage. For more complex queries, you may want to use [Galaxy](#) or our [public MySQL server](#). To examine the biological function of your set through annotation enrichments, send the data to [GREAT](#). Send data to [GenomeSpace](#) for use with diverse computational tools. Refer to the [Credits](#) page for the list of contributors and usage restrictions associated with these data. All tables can be downloaded in their entirety from the [Sequence and Annotation Downloads](#) page.

clade: Mammal **genome:** Human **assembly:** Dec. 2013 (GRCh38/hg38)

group: Repeats **track:** Simple Repeats add custom tracks track hubs

table: simpleRepeat describe table schema

region: genome position chr17 lookup define regions

identifiers (names/accessions): paste list upload list

filter: edit clear

intersection: create

correlation: create

output format: all fields from selected table Send output to ☐ [Galaxy](#) ☐ [GREAT](#) ☐ [GenomeSpace](#)

output file: (leave blank to keep output in browser)

file type returned: ☒ plain text ☐ gzip compressed

get output summary/statistics

To reset **all** user cart settings (including custom tracks), [click here](#).

output



```
#filter: simpleRepeat.copyNum > 10
#bin chrom chromStart chromEnd name period copyNum consensusSize perMatch perIndel score A C G T entropy sequence
585 chr17 77394 77443 trf 2 24.5 2 100 0 98 48 0 0 51 1 TA
585 chr17 80306 80347 trf 3 13.7 3 84 0 55 73 24 0 2 0.96 AAC
585 chr17 92009 92041 trf 2 16 2 93 0 55 53 46 0 0 1 AC
585 chr17 95186 95281 trf 4 23 4 72 10 66 51 0 48 0 1 AAGG
585 chr17 95202 95359 trf 16 10.2 15 72 15 65 43 0 56 0 0.99 AAGGAGGGAGGAGGG
585 chr17 95202 95331 trf 8 16.1 8 76 13 97 45 0 54 0 0.99 AAGGAGGG
585 chr17 95293 95350 trf 4 14.2 4 84 0 69 33 0 66 0 0.92 GAGG
585 chr17 107905 107932 trf 1 27 1 100 0 54 100 0 0 0 0 A
585 chr17 113093 113583 trf 6 79.8 6 85 7 560 33 48 3 14 1.6 ACCCTA
585 chr17 116853 118135 trf 63 20.3 63 97 0 2357 26 7 32 32 1.84
TAGGGCATGGTGGATATTATTACATTAGTATTGGAAGCGATGGTGGACTAGATCAGTGA
585 chr17 122434 128137 trf 114 50 114 98 0 10848 40 16 17 26 1.9
ATTGAGGTTCTCTACTATTACAGCGAGAAGTTACAGATAAAAAATAGGGAAGCCTAGAAATGATCTCTGTCATAATGAGTCAGAAATATATATACAACGTAAGTATAAACTCAC
73 chr17 130917 131507 trf 49 12 49 96 0 1054 36 22 19 21 1.95 AGACCGGAACCCCTAAAAATGATAGTTACTGAGAATACTGCTAATGCCCTA
586 chr17 136264 136435 trf 6 27.5 6 90 8 247 30 53 0 15 1.43 CCTAAC
586 chr17 140598 141646 trf 38 26.9 39 85 8 849 17 13 53 16 1.73 AGGGGACCTGGTGAGGGGACCTGGTGAGGGGACTTGGTG
586 chr17 140598 141628 trf 13 79.5 13 75 17 884 17 13 53 16 1.73 AGGGGACCTGGTG
586 chr17 166599 168758 trf 48 45 48 98 0 4129 23 21 24 31 1.98 ACTGTTCTTGAGGTAGTGAATAACGTCTCACGAGATCTCATGGTCTGAC
586 chr17 170743 172019 trf 32 39.2 32 95 2 2221 19 28 48 3 1.64 CAGAGGGGAGCACCAGGGGCGGGCCAGGCTG
586 chr17 184579 184745 trf 15 11.1 15 78 0 179 22 31 21 24 1.98 CCTTCTGGTGCAAA
586 chr17 187981 188007 trf 1 26 1 100 0 52 0 0 0 100 0 T
586 chr17 190382 190598 trf 4 53.8 4 82 9 244 70 0 28 0 0.91 AAAG
586 chr17 191539 192713 trf 27 44.4 27 94 3 1860 9 57 20 11 1.61 CCCCCGACAGTCACCCGTTCCCCCGC
586 chr17 198309 198680 trf 34 10.9 34 86 3 439 24 33 27 14 1.94 CACAGCAGGATCCCGACTCTGGAACCCAGGCGTT
586 chr17 206415 207086 trf 42 16 42 89 0 991 21 17 34 26 1.95 GGGGAAACGTGATTGGGCGAGTGTTTCTTACGGCAGCTCGAT
586 chr17 215908 216288 trf 33 11.5 33 95 0 672 12 44 26 16 1.83 ACCCCACCCACATGGCTGCCGGGCTCTGGGCTG
586 chr17 216952 218561 trf 83 19 83 88 3 2320 22 28 20 28 1.98
CAGGACCCCCAAGCGATTTCATCCCATCTGGGGCAGTTATTACAGAGCCCTCTTCTGCACTAAAAATTGGCCTCGCTGAAAT
586 chr17 219192 219217 trf 1 25 1 100 0 50 0 0 0 100 0 T
586 chr17 220089 222105 trf 45 44.8 45 95 0 3483 32 34 18 14 1.91 GCACAACAGCTCTGAGGGCTCCACTCACTGAGACAAATAGACCCAA
586 chr17 232829 232887 trf 2 29 2 100 0 116 0 0 50 50 1 GT
586 chr17 233824 234425 trf 37 16.1 37 95 1 1040 21 36 24 17 1.94 GGGAGCGGCTACTTACCCGTACCAACTTCCCCGAGCA
586 chr17 235082 236075 trf 50 19.8 50 88 2 1321 23 25 34 15 1.95 CACTAACAAAGACGGGTCCAGGGTTCAAAGCTGGGGTCGGCCGAGGCTCTG
586 chr17 238137 238210 trf 4 17.8 4 78 10 58 64 0 35 0 0.94 AGAA
586 chr17 241055 241100 trf 3 15 3 100 0 90 66 0 0 33 0.92 AAT
586 chr17 242245 244307 trf 35 59.9 34 91 4 3002 17 33 5 43 1.73 TGATTACCTTCCTCTATTGATTACCTTCCTCTAT
586 chr17 242245 244307 trf 17 119.9 17 92 4 3002 17 33 5 43 1.73 TGATTACCTTCCTCTAT
586 chr17 242245 244307 trf 17 119.9 17 92 4 3002 17 33 5 43 1.73 TGATTACCTTCCTCTAT
```

Simple Repeats (simpleRepeat) Summary Statistics

item count	32,093
item bases	6,820,796 (8.23%)
item total	16,264,416 (19.61%)
smallest item	25
average item	507
biggest item	132,210
smallest score	50
average score	385
biggest score	78,580

Region and Timing Statistics

region	chr17:1-83257441
bases in region	83,257,441
bases in gaps	336,725
load time	0.12
calculation time	0.02
free memory time	0.00
filter	off
intersection	off

file type returned: ☒ plain text ☐ gzip compressed

get output summary/statistics

To reset **all** user cart settings (including custom tracks), [click here](#).



内容提要

- UCSC Genome Browser 简介
- Genome Browser的使用
- Table Browser 的使用
- Download genome annotation

Genome Data Download



[Home](#) [Genomes](#) [Genome Browser](#) [Tools](#) [Mirrors](#) [Downloads](#) [My Data](#) [Help](#) [About Us](#)

Table Browser

Use this program to retrieve the data associated with a track in text format. For a description of the controls in this form, the [Table Browser](#) for a description of the controls in this form, the [Usage](#) page. For more complex queries, you may want to use [Galaxy](#) or [GenomeSpace](#) for use with diverse computational tools. Refer to the [Sequence and Annotation Downloads](#) page.

clade: **genome:** **assembly:**

group: **track:**

table:

region: ☐ genome ☒ position

identifiers (names/accessions):

[Genome Data](#)
[Source Code](#)
[Genome Browser Store](#)
[Utilities](#)
[FTP](#)
[MySQL Access](#)

Sequence and Annotation Downloads

This page contains links to sequence and annotation data downloads for the genome assemblies featured in the UCSC Genome Browser. Table downloads are also available via the Genome Browser [FTP server](#). For quick access to the most recent assembly of each genome, see the [current genomes](#) directory. This directory may be useful to individuals with automated scripts that must always reference the most recent assembly.

To view the current descriptions and formats of the tables in the annotation database, use the "describe table schema" button in the Table Browser. The [Description of the annotation database](#) page (no longer maintained) also provides descriptions of selected tables in the database.

All tables in the Genome Browser are freely usable for any purpose except as indicated in the README.txt files in the download directories. To view restrictions specific to a particular data set, click on the corresponding download link and review the README text. These data were contributed by many researchers, as listed on the Genome Browser [credits](#) page. Please acknowledge the contributor(s) of the data you use.

VERTEBRATES - Complete annotation sets

Human	Gorilla	Platypus
Alpaca	Guinea pig	Rabbit
American alligator	Hedgehog	Rat
Armadillo	Horse	Rhesus
Atlantic cod	Kangaroo rat	Rock hyrax
Baboon	Lamprey	Sheep
Bonobo	Lizard	Shrew
Brown kiwi	Manatee	Sloth
Budgerigar	Marmoset	Squirrel
Bushbaby	Medaka	Squirrel monkey
Cat	Medium ground finch	Stickleback
Chicken	Megabat	Tarsier
Chimpanzee	Microbat	Tasmanian devil
Chinese hamster	Minke whale	Tenrec
Coelacanth	Mouse	Tetraodon
Cow	Mouse lemur	Tree shrew
Crab-eating macaque	Naked mole-rat	Turkey
Dog	Nile tilapia	Wallaby
Dolphin	Opossum	White rhinoceros

以Human基因组信息下载为例

Human Genome

Dec. 2013 (hg38, GRCh38)

- [Full data set](#)
- [Data set by chromosome](#)
- [Annotation database](#)
- [SNP146-masked FASTA files](#)
- [SNP144-masked FASTA files](#)
- [SNP142-masked FASTA files](#)
- [SNP141-masked FASTA files](#)
- [LiftOver files](#)
- Pairwise Alignments
 - [Human self alignments](#)
 - [Human/Chimp \(panTro4\)](#)
 - [Human/Bonobo \(panPan1\)](#)
 - [Human/Rhesus \(rheMac8\)](#)
 - [Human/Rhesus \(rheMac3\)](#)
 - [Human/Crab-eating macaque \(macFas5\)](#)
 - [Human/Tarsier \(tarSyr2\)](#)
 - [Human/Mouse Lemur \(micMur2\)](#)
 - [Human/Gorilla \(gorGor4\)](#)
 - [Human/Mouse \(mm10\)](#)
 - [Human/Rat \(rn5\)](#)
 - [Human/Cow \(bosTau8\)](#)
 - [Human/Cat \(felCat8\)](#)
 - [Human/Dog \(canFam3\)](#)
 - [Human/Opossum \(monDom5\)](#)
 - [Human/X. tropicalis \(xenTro7\)](#)

• Multiple Alignments

- [Multiple alignments of 99 vertebrate genomes with Human](#)
- [Conservation scores for alignments of 99 vertebrate genomes with Human](#)
- [Basewise conservation scores \(phyloP\) of 99 vertebrate genomes with Human](#)
- [FASTA alignments of 99 vertebrate genomes with Human for CDS regions](#)
- [Multiple alignments of 19 mammalian \(16 primate\) genomes with Human](#)
- [Conservation scores for alignments of 19 mammalian \(16 primate\) genomes with Human](#)
- [Basewise conservation scores \(phyloP\) of 19 mammalian \(16 primate\) genomes with Human](#)
- [FASTA alignments of 19 mammalian \(16 primate\) genomes with Human for CDS regions](#)
- [Multiple alignments of 6 vertebrate genomes with Human](#)
- [Conservation scores for alignments of 6 vertebrate genomes with Human](#)
- [Basewise conservation scores \(phyloP\) of 6 vertebrate genomes with Human](#)
- [FASTA alignments of 6 vertebrate genomes with Human for CDS regions](#)
- [GRCh38 Patch 6 - Full data set](#)
- [GRCh38 Patch 3 - Full data set](#)
- [GRCh38 Patch 2 - Full data set](#)

总结

- UCSC可通过检索指定位置，指定物种的基因信息
- 涵盖了基因序列、基因结构、调控区间、表观遗传学、遗传多样性等多种信息
- 同时可以下载特定区域或者全基因组信息



thanks